

GEMINI_API_INTEGRATION_REQUIREMENTS.md

Gemini API Integration Requirements

Purpose

Gemini is the primary intelligence engine powering GSpaceAi.

Gemini is responsible for:

1. Conducting the audit conversation.
2. Extracting structured business information.
3. Building the software inventory.
4. Identifying consolidation opportunities.
5. Identifying automation opportunities.
6. Generating report content.
7. Creating recommendations.
8. Creating implementation guidance.
9. Calculating inputs for scoring and savings estimates.

Gemini should not directly control workflow state.

Gemini should provide intelligence.

The application should control workflow progression.

Core Architecture Principle

Gemini handles reasoning.

The application handles state.

Never rely on Gemini responses to determine:

- payment status

- report unlock status
- workflow stage
- customer permissions
- purchased products

Those must be controlled by application logic.

Preferred Gemini Model

Use the current production Gemini model available at build time.

The implementation should allow model changes through environment variables.

Example:

GEMINI_MODEL=gemini-2.5-pro

Do not hardcode model names.

Gemini Responsibilities

Gemini should support:

Audit Conversation

Generate conversational responses.

Ask follow-up questions.

Discover:

- business type
- services
- software stack
- Google Workspace usage
- workflows
- manual tasks
- bottlenecks

- reporting gaps
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Software Inventory Extraction

Gemini should identify:

- software names
 - categories
 - estimated costs
 - use cases
 - replacement opportunities
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Consolidation Analysis

Gemini should identify:

- software overlap
 - duplicate functionality
 - Google Workspace opportunities
 - consolidation opportunities
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Report Content Generation

Gemini should generate:

- executive summaries
 - findings
 - recommendations
 - implementation guidance
 - SOP content
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Structured Output Requirement

Whenever possible, Gemini should return structured JSON.

Do not rely on long unstructured responses for application logic.

Preferred:

```
{
  "softwareInventory": [
    {
      "name": "Calendly",
      "category": "Scheduling",
      "replacementPotential": "High"
    }
  ]
}
```

Avoid:

Long narrative responses that require extensive parsing.

Audit Data Extraction

After each user response, Gemini should update:

```
{
  "softwareInventory": [],
  "workflowInventory": [],
  "manualTasks": [],
  "automationOpportunities": [],
  "consolidationOpportunities": [],
  "risks": []
}
```

The application should merge these updates into state.

Dual Response Pattern

For audit conversations, Gemini should generate:

1. Customer-facing response

2. Structured data extraction

Example:

```
{
  "customerResponse": "That helps. What software do you currently use to schedule appointments?",
  "extractedData": {
    "softwareInventory": []
  }
}
```

This keeps conversation and application data synchronized.

Google Workspace Expertise Requirements

Gemini should possess strong awareness of:

- Google Workspace
- Gmail
- Google Calendar
- Google Drive
- Google Docs
- Google Sheets
- Google Forms
- Google Sites
- Google Apps Script
- Gemini
- Looker Studio

Gemini should understand how these products work together.

Apps Script Awareness

Gemini should actively look for opportunities where Apps Script could:

- automate emails
- automate reminders
- automate approvals
- automate data movement
- automate document creation
- automate reporting
- automate customer follow-ups
- automate notifications

Apps Script opportunities should be highlighted in paid reports.

Gemini Awareness

Gemini should actively identify opportunities where Gemini itself could help.

Examples:

- email drafting
 - proposal generation
 - SOP generation
 - data summarization
 - customer communication
 - workflow assistance
 - reporting insights
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Tool Classification Framework

Every significant tool discovered should be classified as:

- Keep
- Replace
- Consolidate
- Automate
- Investigate

Gemini should explain why.

Report Generation Rules

Gemini should generate report sections independently.

Example:

Generate:

- Executive Summary
- Software Stack Review
- Recommendations
- Roadmap

as separate sections.

Avoid generating entire reports as one giant response.

This improves formatting consistency.

Hallucination Prevention

Gemini must never:

- invent software costs
- invent integrations
- invent savings guarantees
- invent business processes

When information is unknown:

Use:

"Based on the information provided..."

"Estimated..."

"Potential opportunity..."

Never state assumptions as facts.

Savings Estimate Guidance

Savings should be presented as:

Potential Savings

Estimated Replaceable Spend

Estimated Annual Savings

Never:

Guaranteed Savings

Exact Savings

Guaranteed Cost Reduction

Future Readiness

The Gemini integration should be modular.

Future upgrades may include:

- memory
- customer accounts
- saved audits
- implementation generation
- direct Apps Script generation
- direct Workspace deployment

The architecture should allow Gemini capabilities to expand without major refactoring.

Final Rule

Gemini is the consultant.

The application is the operator.

Gemini should generate intelligence.

The application should control permissions, payments, workflow progression, report gating, and customer state.